

Computational Physics: Materials Science

Exercise 1

Projectile dynamics: analytical vs. numerical integration

Date of delivery:	4 May 2026
Authors:	Stefanus Michael 5976298 Microsystems Engineering

Projectile Dynamics: Analytical vs. Numerical Integration

Numerical integration is at the heart of virtually every computational physics simulation: differential equations that govern a system's evolution are stepped forward in time using a chosen algorithm. The choice of integrator is not merely a technical detail — different schemes differ fundamentally in accuracy, stability, and whether they respect the underlying physics encoded in the Hamiltonian structure of the equations. This exercise uses projectile motion as a controlled test case because its exact analytical solution is known, allowing a rigorous, quantitative comparison of two widely used integration methods: **Explicit Euler** (first-order, non-symplectic) and **Velocity Verlet** (second-order, symplectic). Understanding these trade-offs is directly relevant to molecular dynamics, orbital mechanics, and any long-time simulation where energy conservation matters.

Implementation

Goal and Physical Setup

This exercise simulates the two-dimensional motion of a projectile of mass m launched with initial speed v_0 at angle θ in a uniform gravitational field g . The two integration schemes are compared against the exact analytical solution. An AI assistant (Open-WebUI) was used as a coding support tool; outputs were cross-checked against Claude Sonnet 4.6 and ChatGPT, and all physical interpretation was carried out independently.

Equations of Motion

Without air resistance: $\ddot{x} = 0$, $\ddot{y} = -g$. With zero initial position and $(\dot{x}_0, \dot{y}_0) = (v_0 \cos \theta, v_0 \sin \theta)$ the exact solution is

$$x(t) = v_0 \cos \theta \cdot t, \quad y(t) = v_0 \sin \theta \cdot t - \frac{1}{2}gt^2. \quad (1)$$

This gives flight time $T = 2v_0 \sin \theta / g$, range $R = v_0^2 \sin 2\theta / g$, and maximum height $H = (v_0 \sin \theta)^2 / (2g)$.

Integration Schemes

Explicit Euler (1st order). $\mathbf{r}_{n+1} = \mathbf{r}_n + \mathbf{v}_n \Delta t$, $\mathbf{v}_{n+1} = \mathbf{v}_n + \mathbf{a}_n \Delta t$. Using only old-step information makes this *non-symplectic*, causing a systematic upward energy drift that grows linearly in time.

Velocity Verlet (2nd order, symplectic). $\mathbf{r}_{n+1} = \mathbf{r}_n + \mathbf{v}_n \Delta t + \frac{1}{2}\mathbf{a}_n \Delta t^2$, $\mathbf{v}_{n+1} = \mathbf{v}_n + \frac{1}{2}(\mathbf{a}_n + \mathbf{a}_{n+1})\Delta t$. For velocity-dependent forces, $\mathbf{a}_{n+1}$ uses a predictor step $\mathbf{v}^* = \mathbf{v}_n + \mathbf{a}_n \Delta t$ (predictor–corrector).

Code Structure

The Python script is modular: **pluggable acceleration functions** (`acceleration_no_drag`, `acceleration_linear_drag`, `acceleration_quadratic_drag`) are passed to the **integrators**, so adding a new force model requires no integrator changes. **Linear ground**



interpolation extracts accurate flight times, and five **analysis functions** each produce a figure and a console summary. All physical parameters (m , v_0 , θ , g , Δt) are collected in a single **params** dictionary at the top of the script for reproducibility. Base parameters: $m = 1$ kg, $v_0 = 50$ m/s, $\theta = 45$, $g = 9.807$ m/s², $\Delta t = 0.01$ s.

Analysis

1. Trajectory and Time of Flight

Figure 1 shows that Velocity Verlet overlaps the analytical parabola to plotting accuracy, while Euler overshoots slightly due to its first-order truncation error. The position error (panel b) confirms Verlet is orders of magnitude more accurate at the same Δt . Numerically, the range increases from 254.93 m (analytical) to 255.28 m for Euler — an overshoot of 0.35 m — while Verlet matches the analytical value to five significant figures. Table 1 summarises the key quantities.

Table 1: Key trajectory quantities at $\Delta t = 0.01$ s, $v_0 = 50$ m/s, $\theta = 45$, $g = 9.807$ m/s².

	Analytical	Explicit Euler	Velocity Verlet
Flight time (s)	7.2105	7.2205	7.2105
Range (m)	254.93	255.28	254.93
Max. height (m)	63.73	63.91	63.73

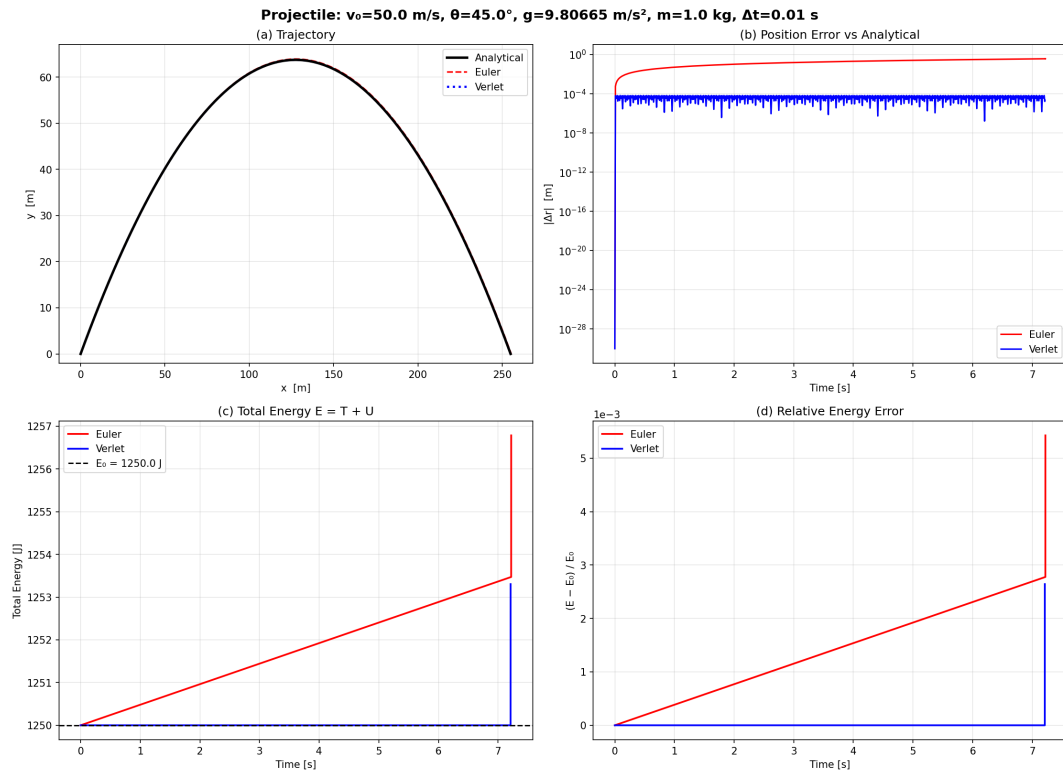


Figure 1: Trajectory and energy overview ($\Delta t = 0.01$ s, no drag). **(a)** Trajectories: Verlet and analytical overlap; Euler overshoots. **(b)** Position error $|\Delta \mathbf{r}|$ on a log scale. **(c)** Total energy $E = T + U$: Euler drifts upward; Verlet is flat. **(d)** Relative energy error confirming Euler's linear growth.



2. Energy Conservation: Euler vs. Velocity Verlet

The total mechanical energy $E = \frac{1}{2}m|\mathbf{v}|^2 + mgy$ must equal $E_0 = \frac{1}{2}mv_0^2 = 1250 \text{ J}$ throughout the flight for a conservative force field.

Why Euler drifts: at each step the old force advances both position and velocity, injecting $\sim \frac{1}{2}mg^2\Delta t^2$ of spurious energy per step. Accumulated over $N = T/\Delta t$ steps, the total drift is $\Delta E \approx \frac{1}{2}mg^2\Delta t T \approx 3.47 \text{ J}$ (estimated); the simulation yields 6.79 J , giving a relative drift $\Delta E/E_0 = 5.43 \times 10^{-3}$. This error grows *linearly in time*, clearly visible as the upward slope in Figure 1c–d.

Why Verlet conserves energy: Velocity Verlet is a *symplectic integrator* that preserves the Hamiltonian structure of the equations of motion. As a consequence, its energy error does *not* drift monotonically; instead it remains *bounded* and oscillates around E_0 with an amplitude proportional to Δt^2 . This is the defining hallmark of symplectic integrators and is fundamentally different from both Euler’s secular drift and true floating-point round-off (which would be at machine epsilon $\sim 10^{-16}$). The observed $\Delta E/E_0 = 2.64 \times 10^{-3}$ at $\Delta t = 0.01 \text{ s}$ is therefore a Verlet discretisation effect, not numerical noise, and it decreases as Δt^2 when the step is reduced (confirmed in Section 3).

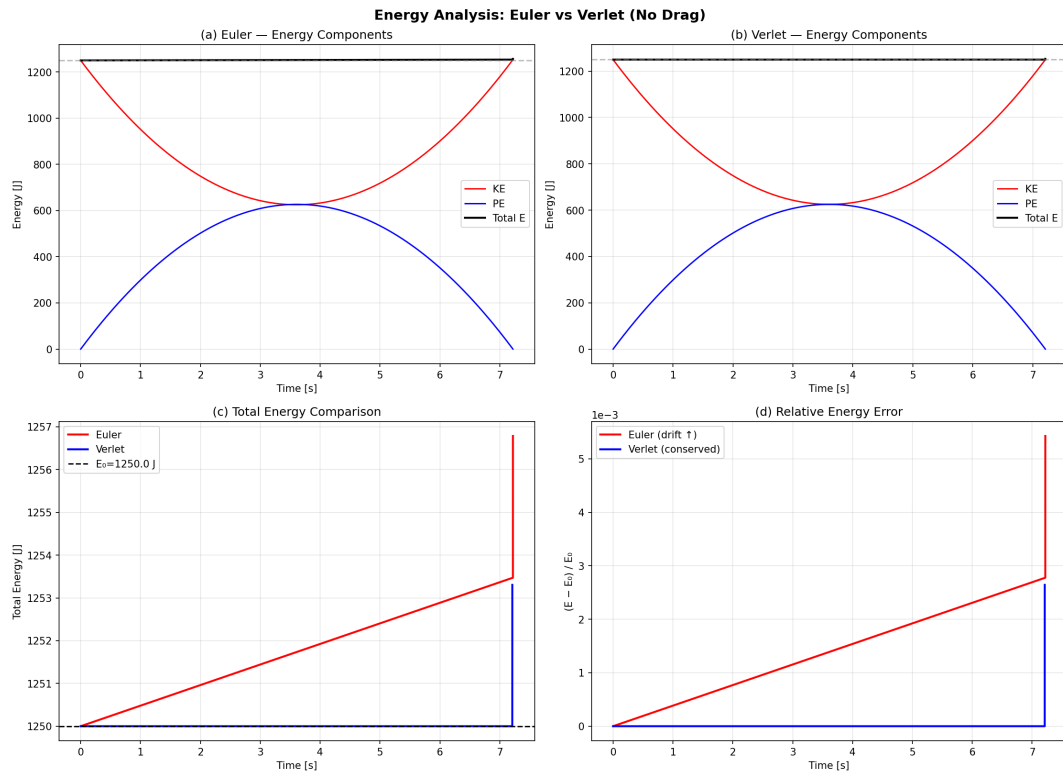


Figure 2: Energy components KE, PE, total E for (a) Euler and (b) Verlet; (c–d) direct comparison and relative error. Euler shows a clear linear drift; Verlet’s error is bounded and oscillatory, consistent with symplectic behaviour.

3. Effect of the Time Step Δt

Figure 3 confirms $\mathcal{O}(\Delta t)$ convergence for Euler and $\mathcal{O}(\Delta t^2)$ for Verlet on log–log axes, consistent with their respective truncation orders. Halving Δt halves Euler’s error but



quarters Verlet's. Verlet achieves the same positional accuracy as Euler with a $\sim 10\times$ larger step (Table 2), making it both more accurate *and* more computationally efficient. The energy error column confirms that Verlet's bounded error also scales as $\mathcal{O}(\Delta t^2)$, validating the symplectic interpretation in Section 2.

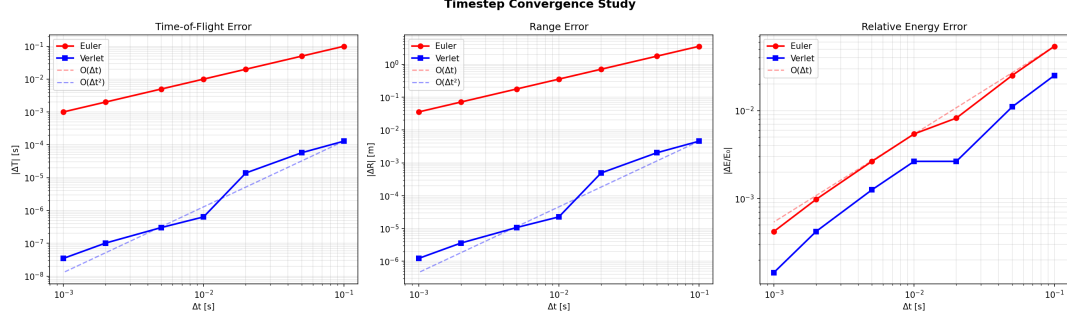


Figure 3: Convergence study ($\Delta t \in [10^{-3}, 10^{-1}]$ s). Dashed lines show $\mathcal{O}(\Delta t)$ and $\mathcal{O}(\Delta t^2)$ reference slopes. Both range error and energy error follow these slopes, confirming the theoretical orders of the two integrators.

Table 2: Range error $|\Delta R|$ [m] and relative energy error $|\Delta E/E_0|$ for selected Δt . Verlet's errors decrease as Δt^2 , confirming symplectic $\mathcal{O}(\Delta t^2)$ scaling.

Δt [s]	$ \Delta R $ [m]		$ \Delta E/E_0 $	
	Euler	Verlet	Euler	Verlet
0.100	3.53	4.6×10^{-3}	5.4×10^{-2}	2.5×10^{-2}
0.010	3.5×10^{-1}	2.3×10^{-5}	5.4×10^{-3}	2.6×10^{-3}
0.001	3.5×10^{-2}	1.2×10^{-6}	4.2×10^{-4}	1.4×10^{-4}

4. Air Resistance

Two drag models are considered. **Linear (Stokes) drag**, $\mathbf{F}_{\text{drag}} = -b\mathbf{v}$, applies at low Reynolds numbers (viscous flow) and modifies the acceleration to $\ddot{x} = -\gamma\dot{x}$, $\ddot{y} = -g - \gamma\dot{y}$ ($\gamma = b/m$), with closed-form solution

$$x(t) = \frac{v_{x0}}{\gamma}(1 - e^{-\gamma t}), \quad y(t) = \frac{1}{\gamma} \left(v_{y0} + \frac{g}{\gamma} \right) (1 - e^{-\gamma t}) - \frac{g}{\gamma} t. \quad (2)$$

Quadratic (Newtonian) drag, $\mathbf{F}_{\text{drag}} = -\frac{1}{2}C_d\rho A|\mathbf{v}|\mathbf{v}$, is more realistic for macroscopic projectiles at typical speeds (high Reynolds numbers) where pressure drag dominates. No closed-form solution exists for the two-dimensional quadratic case, so numerical integration is the only option; the modular code handles this by passing `acceleration_quadratic_drag` to the same integrators without modification.

In both models the trajectory becomes asymmetric: drag decelerates the projectile throughout, but the horizontal deceleration is stronger on the way up (where v_y adds to the speed magnitude) than on the way down, producing a steeper descent angle than ascent. Even modest linear drag ($b = 0.1$ kg/s) cuts the range by $\approx 34\%$; at $b = 1.0$ kg/s the range is reduced by $\approx 86\%$ to only 35 m.



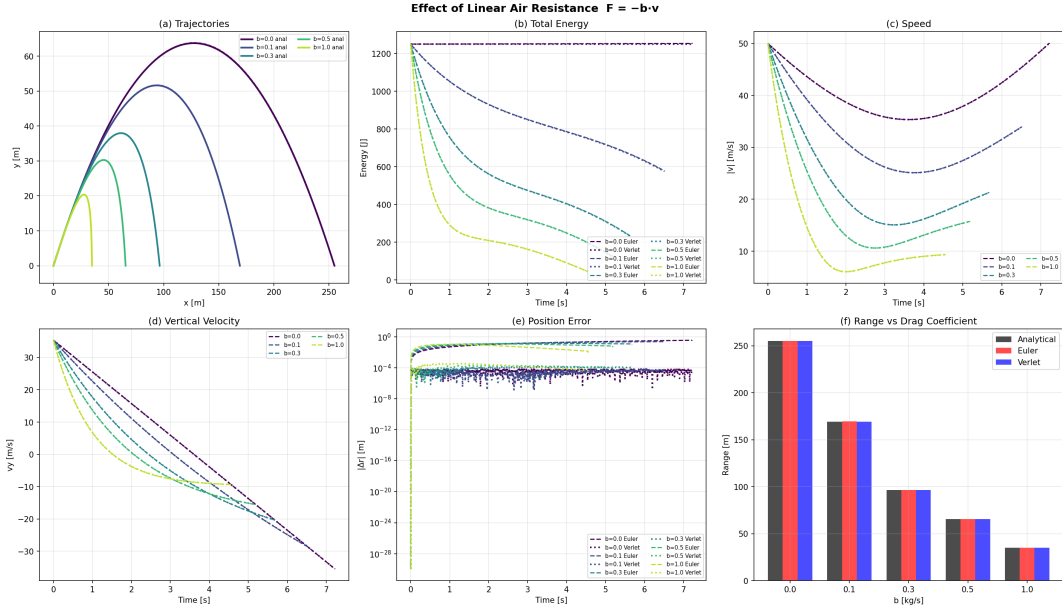


Figure 4: Linear drag for $b \in \{0.0, 0.1, 0.3, 0.5, 1.0\}$ kg/s. **(a)** Trajectories shorten and become asymmetric. **(b)** Energy decreases monotonically ($dE/dt = -b|\mathbf{v}|^2 \leq 0$). **(c)** Speed: landing speed equals launch speed only for $b = 0$. **(d)** v_y : apex shifts earlier with drag. **(e)** Verlet position error vs. drag analytical solution. **(f)** Range vs. b : all three methods agree.

Table 3: Analytical flight time and range for selected linear drag coefficients ($v_0 = 50$ m/s, $\theta = 45^\circ$, $m = 1$ kg).

b [kg/s]	γ [s ⁻¹]	T [s]	R [m]
0.0	0.00	7.21	254.9
0.1	0.10	6.51	169.2
0.5	0.50	5.19	65.4
1.0	1.00	4.56	35.0

5. Parameter Variation: What Happens on Mars?

Mars has $g_{\text{Mars}} = 3.721 \text{ m/s}^2 \approx 0.38 g_{\oplus}$, giving a scaling ratio $g_{\oplus}/g_{\text{Mars}} \approx 2.64$. From the analytical formulae, both flight time and range scale as $1/g$, so all kinematic quantities should increase by this factor on Mars. The simulation confirms this: range increases from 254.9m (Earth) to 671.9m (Mars), and flight time from 7.21s to 19.00s, both matching the $2.64\times$ prediction to three significant figures (Table 4, Figure 5).

The initial kinetic energy $E_0 = \frac{1}{2}mv_0^2 = 1250 \text{ J}$ is identical on both planets (it depends only on v_0 , not g). Verlet conserves it to its characteristic bounded-oscillation precision on both (Figure 5d), confirming that symplecticity is a property of the integrator and is independent of the gravitational constant.

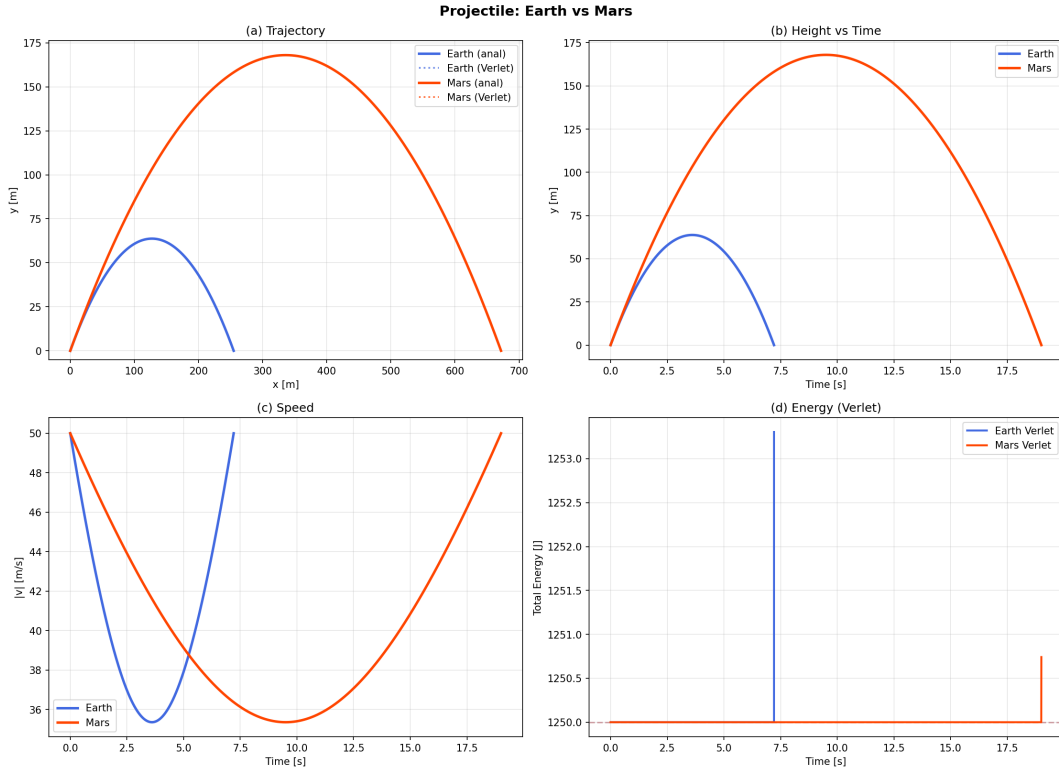


Figure 5: Earth (blue) vs. Mars (red), $v_0 = 50 \text{ m/s}$, $\theta = 45$, no drag (solid: analytical; dotted: Verlet). **(a)** Mars range $\approx 2.64\times$ larger (672m vs. 255m). **(b)** Flight time $\approx 2.64\times$ longer (19.0s vs. 7.2s). **(c)** Apex speed identical; duration longer on Mars. **(d)** Verlet conserves E_0 to bounded-oscillation precision on both planets.

Table 4: Analytical results on Earth and Mars ($v_0 = 50 \text{ m/s}$, $\theta = 45$). All quantities scale by $g_{\oplus}/g_{\text{Mars}} \approx 2.64$.

Planet	g [m/s ²]	T [s]	R [m]	H [m]
Earth	9.807	7.21	254.9	63.7
Mars	3.721	19.00	671.9	168.0
Ratio	2.636	2.636	2.636	2.636

Summary

- **Velocity Verlet** is strongly preferred over Explicit Euler: it is 2nd-order ($\mathcal{O}(\Delta t^2)$), symplectic, and its energy error is *bounded* rather than drifting — achieving the same accuracy as Euler with a $\sim 10\times$ larger time step.
- **Explicit Euler** is 1st-order ($\mathcal{O}(\Delta t)$) and non-symplectic, producing a *linear-in-time* energy drift with measurable errors in range, flight time, and height.
- Verlet’s energy error ($\Delta E/E_0 \propto \Delta t^2$) is a bounded symplectic oscillation, *not* floating-point round-off; it decreases quadratically as the step is refined, as confirmed by the convergence study.
- **Air drag** (linear or quadratic) dissipates energy, produces asymmetric trajectories with a steeper descent than ascent, and dramatically reduces range — by $\approx 34\%$ for $b = 0.1 \text{ kg/s}$ and $\approx 86\%$ for $b = 1 \text{ kg/s}$. The modular code handles both models without modifying the integrators.

- On **Mars** ($g \approx 0.38 g_{\oplus}$) all kinematic quantities (range, flight time, height) scale by $g_{\oplus}/g_{\text{Mars}} \approx 2.64$, consistent with the $1/g$ analytical prediction; Verlet's symplectic conservation is unaffected by the change in g .

